Cap-Selected Honeycomb Amplitudes: Marked-Arch Budgets, Confluent Jets, and Certified Height Truncation

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Abstract

We analyze the finite-circumference interfaces in the source manuscript *Cap-selected amplitudes and triangle chords for honeycomb walks*. The selected amplitude is a boundary-paired coefficient, not an eigenvalue or an arbitrary normalized eigenvector overlap. Keeping the source's marked completion argument, we replace its budget of N exceptional rows by the actual number of initial matching arches, at most $\lfloor N/2 \rfloor$. We retain the resulting negative-binomial majorant explicitly and prove a finite-height amplitude enclosure with separate cap, middle-transfer, and omitted-convolution errors. An unequal cap-depth certificate and an exact rational example make the stopping rule concrete. We also explain why collision jets and the physical port/edge conventions cannot be discarded. The full residue compactification, real-root selection, circumference asymptotics, and planar projection are source inputs, not independently certified global results of this report. No uniform circumference spectral gap or new honeycomb critical exponent is claimed. Contribution assessment: C3/5, a source-specific marked-arch refinement and explicit cap-amplitude truncation certificate are proved, conditional on the stated fixed-circumference transfer bounds.

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1 The Model and the Selected Quantity

Let $l = \pi/8$ and $x = (2\cos l)^{-1} = (2+\sqrt{2})^{-1/2}$. The dual of the unit triangular tiling is the honeycomb graph. A port is the midpoint of a triangle side. A port-to-port self-avoiding path visits m triangle centres and has weight x^m. Deleting its two boundary half-edges leaves $m-1$ centre-to-centre edges. Both endpoint orders are counted in the source's triangle law.

For circumference N, the horizontal translation is by N, and the row index increases under translation by $\mathrm{e}^{i\pi/3}$. The positive two-tip partition $\mathcal{G}_N(H)$ counts compatible polygons that either wind between cuts $0, H$, or are contractible and enclose exactly one tip. Each surviving polygon has activity $2x^{|P|}$; the empty family contributes one. If λ_N is the winding-only row growth, the source's amplitude is

$$\mathcal{A}_N = \lim_{H\to\infty} \lambda_N^{-H}\mathcal{G}_N(H).$$

This is a fixed-N limit. The source subsequently claims $\mathcal{A}_N = N^{1/6+o(1)}$, planar nesting $Z(r) = r^{1/12+o(1)}$, and a mean triangle chord length $R^{4/3+o(1)}$. The latter uses all lengths at critical fugacity, with a fixed macroscopic diameter cutoff [1]. These global conclusions remain attributed to that manuscript.

There are two different finite state spaces. The spin row $T^{K^{-1}}(u)$ acts on $(\mathbb{C}^3)^{\otimes N}$, with spin labels $0, +1, -1$. The annular diagram row M acts on the finite set of noncrossing annular matching states, including their homotopies and the empty state 0. No strand goes to infinity. The orientation comparison equates specified capped scalar contractions; it does not identify the two state spaces or make the spin row Hermitian.

The right disk cap Ψ has twist i, and the left cap Λ uses reversed sites, transposition and spin negation, hence twist $-i$. Their empty coefficients are one and their bilinear pairing is $\Lambda\Psi = 1$. The transposes in this algebraic construction are not conjugate transposes. In particular the cup contraction can vanish although the cup tensor is nonzero. We will not use that pairing as a positive Hilbert norm.

2 What the Source Establishes at Each Interface

All nineteen cached text files, including the included higher-jet module, were freshly displayed in bounded, untruncated intervals. Reading them does not amount to an independent certification of all their arguments. The following map records the dependencies retained in this report.

Source module	Mechanism and retained obligation
`local`, `vacuum`	Ordered R-matrices, bilinear cups, fusion, generic disk caps and their denominator. Absence of a generic equal-site divisor is not by itself regularity at a simultaneous homogeneous collision.
`residue`, `common-twist`	Independent twists, compactification and boundary valuations, then a holomorphic common-twist trace. Colliding roots contribute finite jets; toric-end exclusions are stated at generic twists.
`site-moments`, `repeated-moments`	Exact cap normalization from the empty-row moment. Two-site fusion removes auxiliary pairs, retaining the cleared-relation ideal and all collision jets, to obtain arbitrary repeated row moments.
`counting`	Explicit density, consecutive-root quantile box, bulk and two edge deficit estimates, a uniform real ℓ^∞ inverse, and fixed-N local analytic continuation.
`physical-amplitude`	Positive ordinary branch, isolated scalar coefficient, primitive annular row, nonempty-block gap, cap-end bounds and first/last-empty-cut factorization. The gap and complex neighborhoods can depend on N.
`asymptotics`	Quantile discrepancy, missing diagonal term, logarithmic harmonic sums and Jacobian determinant. This supplies the claimed 1/6 amplitude power, not our finite certificate.
`cylinder`, `higher-jets`	Positive boundary flux, finite caps, strip convolution on $\ell^2(n)$, a separate ℓ^1 argument, and every fixed winding derivative. No uniform fugacity analytic radius is assumed.
`projection`, `chords`	Positive polymer coefficients, height moments, rotated-cylinder projection, slit identity, macroscopic denominator and mean length. Topology, positivity and the order of limits remain indispensable.

For example the source counting density is

$$\rho(y) = \frac{2\sqrt{3}}{3\pi}\left(\frac{1}{\cosh(8y/3) - 1/2} + \frac{1}{\cosh(8y/3) + 1/2}\right), \qquad \int \rho = 1.$$

With $a_j = N\rho(y_j)$, $f_i = \partial_y T(y_i)$, its actual quantile Jacobian is

$$H_{ij} = \delta_{ij}\frac{f_i}{a_i} - \frac{r(y_i - y_j)}{a_j}.$$

The self term in f_i is subtracted on the diagonal. The source already treats both edge roots and proves a uniform real inverse on its box. We neither claim this as a new result nor turn it into a uniform complex neighborhood.

3 The Marked-Arch Budget

Fix an admissible circumference and physical real parameters for which the source's annular construction applies. Let A be the principal nonempty-state block of M. Fix a submultiplicative induced matrix norm and verified constants

$$0 < a < \lambda, \qquad C_A \geq 1, \qquad \left\|A^j\right\| \leq C_A a^j \quad (j \geq 0), \qquad \|X\| \leq X_*. \tag{1}$$

Here $\lambda > 0$ is the selected leading eigenvalue. The nonnegative matrix X is the exceptional-row majorant: temporarily replace closed-loop weights by one fixed positive upper weight. All comparison constants for cap coefficients, endpoint transition weights and norm conversion are kept separately below. The existence of such constants follows at fixed N from the source gap; their numerical certification is additional work.

Proposition 3.1 (Occupancy-resolved exceptional budget)**.** *In the marked-completion argument of* `physical-amplitude.tex`*, lines 219–246, a bottom occupation of $2k_0$ sites needs exactly k_0 matching arches in its completion. A completed diagram can have at most k_0 rows in which a closed component uses a marked completion arch. In particular, the source's budget N can be replaced by*

$$k = \lfloor N/2 \rfloor.$$

For a fixed initial completion, the bulk majorant is

$$U_m^{(k_0)} = \sum_{r=0}^{\min(k_0,m)} \sum_{\substack{m_0+\cdots+m_r=m-r \\ m_i \geq 0}} A^{m_r} X A^{m_{r-1}} \cdots X A^{m_0}. \tag{2}$$

It obeys the scalar bound

$$\left\| U_m^{(k_0)} \right\| \leq u_m^{(k_0)} := \sum_{r=0}^{\min(k_0,m)} \binom{m}{r} C_A^{r+1} X_*^r a^{m-r}. \tag{3}$$

Proof. The state is a matching, not a collection of open rays. Every initial completion arch joins two distinct occupied sites. Mark these k_0 arches. During row propagation, marks travel with their connected components; joining components unions their marks and creates none. When a component closes it is removed. If it uses a marked arch it therefore consumes at least one of the original marks permanently. Distinct exceptional rows consume disjoint nonempty sets of marks. This proves the budget, even if one row closes several marked components. Winding homotopy does not create extra endpoints or marks.

All other closed components receive the bulk weights already represented by A. The source's orientation argument supplies bounded cap coefficients: it depends on the bounded number of surviving strands, not on treating each row as a new orientation choice. We retain this source input and its fixed-N prefactor. Forgetting propagated marks therefore introduces no extra per-row multiplicity. For r exceptional rows, the ordinary stretches have lengths $m_0, \ldots, m_r$; stars and bars gives $\binom{(m-r)+r}{r} = \binom{m}{r}$. Submultiplicativity and (1) bound each word by $C_A^{r+1} X_*^r a^{m-r}$. Summing proves the assertion. □

This strengthens the source's polynomial prefactor from $(m+1)^N$ to $(m+1)^{\lfloor N/2 \rfloor}$, with different fixed-$N$ constants permitted. It does not change the base a, prove a uniform gap, or reduce the annular state-space dimension. If individual initial occupations are recorded, their own k_0 can be used before summing cap weights; the worst-case bound is unnecessary for a sector with fewer occupied sites.

4 Exact Cap Tail Majorants

Normalize by the selected growth:

$$\alpha = a/\lambda \in (0,1), \quad b = C_A X_*/\lambda, \quad v_m = \lambda^{-m} u_m^{(k)} = C_A \sum_{r=0}^{\min(k,m)} \binom{m}{r} b^r \alpha^{m-r}.$$

All quantities here are real and nonnegative. The following elementary generating functions are classical negative-binomial identities, used with the source's new arch budget.

Lemma 4.1 (Full mass and exact tail)**.** *For $|z| < \alpha^{-1}$,*

$$U(z) := \sum_{m\ge 0} v_m z^m = C_A \sum_{r=0}^{k} \frac{(bz)^r}{(1-\alpha z)^{r+1}}. \tag{4}$$

For every integer $L \ge 0$, the tail is the finite positive expression

$$T_k(L) := \sum_{m\ge L} v_m = C_A \sum_{r=0}^{k} b^r \sum_{j=0}^{\min(r,L)} \binom{L}{j} \frac{\alpha^{L-j}}{(1-\alpha)^{r-j+1}}. \tag{5}$$

Thus no cancellation of nearly equal infinite sums is necessary.

Proof. For $r \ge 0$, differentiating the geometric series r times and dividing by $r!$ gives

$$\sum_{m\ge r} \binom{m}{r} \alpha^{m-r} z^m = z^r (1-\alpha z)^{-r-1}.$$

There are only $k+1$ terms. For the tail write $m = L + n$, use Vandermonde's identity $\binom{L+n}{r} = \sum_j \binom{L}{j}\binom{n}{r-j}$, and sum the nonnegative geometric derivatives in n. The factor is $\alpha^{L-j}(1-\alpha)^{-r+j-1}$. The indices $j \le L, r$ ensure nonnegative powers, including $L < r$. □

The pole order of U at $1/\alpha$ is at most $k+1$. Thus the polynomial factor has a precise combinatorial meaning: exceptional rows may be placed among ordinary stretches. It need not reflect a Jordan block of the physical transfer M.

5 A Finite-Height Amplitude Certificate

The source's first/last-empty-cut decomposition is

$$G_H = C_H + \sum_{\substack{i,j\ge 0 \\ i+j\le H}} E_i^+ Z_{H-i-j} E_j^-, \qquad Z_m = \langle 0|M^m|0\rangle. \tag{6}$$

Here $E_0^\pm = 1$, $C_0 = 0$. The C_H diagrams have no empty cut; E^- ends at the first empty cut and E^+ begins at the last. The two directions need not have the same coefficients. The source's cap-end estimate, with Proposition 3.1, has the following quantitative form when its constants are supplied:

$$|C_m| \le B_C u_m \quad (m \ge 1), \qquad |E_m^\pm| \le B_\pm u_{m-1} \quad (m \ge 1). \tag{7}$$

All finitely many choices of initial completion and bounded endpoint weights are included in $B_C, B_\pm$, not silently set to one.

Suppose also that a certified middle-transfer estimate is available:

$$|\lambda^{-m} Z_m - d| \le B_Z t^m \quad (m \ge 0), \qquad 0 < t < 1. \tag{8}$$

For a simple isolated eigenvalue, $d = \langle 0|\Pi_\lambda|0\rangle$. The bound at $m = 0$ is required as well; if necessary enlarge B_Z. This scalar premise can be supplied by a full finite-state resolvent or power bound. Checking finitely many powers without a tail argument does not establish it.

Define

$$\beta_\pm = B_\pm/\lambda, \qquad F_\pm(z) = 1 + \beta_\pm z U(z), \qquad K(z) = F_+(z)F_-(z), \tag{9}$$

and let $K_j = [z^j]K(z) \ge 0$. These are majorants; they are not the actual cap functions.

Theorem 5.1 (Explicit three-part error)**.** *Under* (1), (6), (7) *and* (8), *set*

$$S_{\pm} = \sum_{m \geq 0} \lambda^{-m} E_m^{\pm}, \qquad c = dS_+S_-.$$

The sums converge absolutely and, for every integer $H \geq 0$,

$$\left|\lambda^{-H} G_H - c\right| \leq B_C v_H + B_Z[z^H]\frac{K(z)}{1-tz} + |d|\left(K(1) - \sum_{j=0}^{H} K_j\right). \tag{10}$$

The first term may be omitted at $H = 0$. *Each term on the right is computable from the declared constants by finite rational operations when those constants are rational. It tends to zero. Consequently the amplitude limit is* c.

Proof. Put $e_m^{\pm} = \lambda^{-m} E_m^{\pm}$ and $z_m = \lambda^{-m} Z_m = d + r_m$. By (7),

$$|e_0^{\pm}| = 1, \qquad |e_m^{\pm}| \leq \beta_{\pm} v_{m-1} \quad (m \geq 1).$$

Thus their absolute generating series are coefficientwise bounded by $F_{\pm}$, and their full absolute sums by $F_{\pm}(1) < \infty$. Substitution in (6) gives exactly

$$\lambda^{-H} G_H - c = \lambda^{-H} C_H + \sum_{i+j \leq H} e_i^{+} r_{H-i-j} e_j^{-} - d \sum_{i+j > H} e_i^{+} e_j^{-}.$$

The three absolute bounds are respectively those displayed in (10). In particular, the last term is a triangular convolution tail; replacing it by only one endpoint tail would miss pairs with both individual depths at most H.

Since v_m is a polynomial of degree at most k in m, times α^m, the first term vanishes. The coefficients of K are summable. For the middle term, split its convolution at an index J: the finite prefix tends to zero as $H \to \infty$; the remaining sum is bounded by $\sum_{j>J} K_j$. First take $H \to \infty$, then $J \to \infty$. The final term is the tail of that same convergent series. □

Corollary 5.2 (Polynomial order is retained)**.** *For fixed certified data, with* $q = \max(\alpha, t) < 1$, *the right side of* (10) *is*

$$O((H+1)^{2k+2} q^H).$$

The exact coefficient formula is usually preferable to this coarser summary. If $t > \alpha$, *its middle term is at most* $B_Z t^H K(1/t)$, *since* $K(1/t) < \infty$.

Proof. Convolution of two sequences bounded by $C(m+1)^k \alpha^m$, allowing the harmless shifted indices in $F_{\pm}$, is bounded by $C'(m+1)^{2k+1}\alpha^m$. Summing once more against t^{H-m} costs at most $H+1$. For the tail, write $m = H+1+n$ and use $(H+2+n)^{2k+1} \leq (H+2)^{2k+1}(n+1)^{2k+1}$; the remaining weighted series is finite. The last assertion follows by enlarging the finite sum of $K_j t^{-j}$ to its full sum. □

The certificate supplies a stopping rule: evaluate its right side until it is below a prescribed absolute tolerance. This is conditional on validated input constants and cap evaluations. It is not a claim that constructing the full annular transfer is inexpensive, nor that the source provides those constants uniformly in N. In particular $t_N \uparrow 1$ can defeat every preassigned height schedule $H(N)$.

5.1 Unequal cap depths and a finite middle estimate

Computing the two caps to the same depth is unnecessary. Write

$$s_{\pm,L} = \sum_{m=0}^{L} e_m^{\pm}, \qquad \varepsilon_{\pm,L} = \beta_{\pm} T_k(L).$$

The shift in the tail index is important: $\sum_{m>L} \beta_{\pm} v_{m-1} = \beta_{\pm} T_k(L)$.

Proposition 5.3 (Boundary-paired truncation)**.** *For any integers $L_+, L_-, J \geq 0$, let*

$$\widehat{d} = \lambda^{-J} Z_J, \quad \delta_d = B_Z t^J, \quad \widehat{c} = \widehat{d}\, s_{+,L_+} s_{-,L_-}.$$

Put $s_{\pm} = s_{\pm,L_\pm}$, $\varepsilon_{\pm} = \varepsilon_{\pm,L_\pm}$. Then

$$|c - \widehat{c}| \leq \delta_d |s_+ s_-| + (|\widehat{d}| + \delta_d)(\varepsilon_+ |s_-| + \varepsilon_- |s_+| + \varepsilon_+ \varepsilon_-). \tag{11}$$

Certified arithmetic errors in each $s_\pm$ can be added to $\varepsilon_\pm$, and an arithmetic error in $\widehat{d}$ to δ_d. The product-error term must be retained.

Proof. Write $S_\pm = s_\pm + \Delta_\pm$, where $|\Delta_\pm| \leq \varepsilon_\pm$, and $d = \widehat{d} + \Delta_d$, where $|\Delta_d| \leq \delta_d$. Expand $dS_+S_- - \widehat{d}s_+s_-$, grouping first the error in d, and then the three terms in the product of the caps. Use $|d| \leq |\widehat{d}| + \delta_d$. The same argument applies to interval or ball arithmetic errors. □

If an enclosure has centre $\widehat{c}$ and radius ε_c with $\widehat{c} > \varepsilon_c$ in a real positive setting, it certifies nonvanishing and gives the relative error $\varepsilon_c/(\widehat{c} - \varepsilon_c)$. Otherwise it supplies only an absolute enclosure; physical positivity does not make an inaccurate calculation relatively accurate.

6 Worked Finite Model and What It Does Not Test

Take $\lambda = 1$, $a = 2/3$, $C_A = 1$, $X_* = 1/6$, $k = 1$, and $B_+ = B_- = 1/6$. Then

$$v_m = (1 + m/4)(2/3)^m, \qquad U(1) = 9/2, \qquad F_+(1) = F_-(1) = 7/4.$$

Set $C_H = 0$ and take the actual cap sequences equal to their majorants. A two-state middle row is

$$M = \begin{pmatrix} 7/9 & 2/9 \\ 4/9 & 5/9 \end{pmatrix}, \qquad Z_J = (M^J)_{00} = \frac{2}{3} + \frac{1}{3}(1/3)^J.$$

Its nonempty block is $5/9 < 2/3$. Thus $d = 2/3$, $B_Z = 1/3$, $t = 1/3$ are valid, and the exact limiting amplitude is

$$c = \frac{2}{3}(7/4)^2 = \frac{49}{24}.$$

The cap tail simplifies without subtraction to

$$\varepsilon_{\pm,L} = (2/3)^L (3/4 + L/8). \tag{12}$$

For equal depth 20 each cap error is below $1/1000$; taking $J = 20$ in (11) gives an amplitude error below $1/400$. These rational inequalities are checked exactly by the owned script. Unequal depths are tested as well.

This is an exact manufactured transfer/convolution model, not an enumeration of the honeycomb annular states. It tests the formulas, normalizations and nonnegative tail accounting. It does not establish the source's λ_N, W/J, edge-root constants or physical amplitude for any particular honeycomb circumference.

7 Supplying the Middle Bound Without a Spectral Guess

Here is one fully finite sufficient route in the positive winding regime. It is the classical stochastic overlap contraction, applied to the actual annular row, not to a compression of it.

Proposition 7.1 (Perron-scaled overlap certificate)**.** *Let the full finite nonnegative annular matrix satisfy $Mv = \lambda v$ with $v > 0$, $\lambda > 0$, and put*

$$P = \lambda^{-1} \operatorname{diag}(v)^{-1} M \operatorname{diag}(v).$$

Thus every row of P sums to one. Suppose an integer $b_0 \geq 1$ and a certified number $0 < \delta < 1$ satisfy

$$\max_{i,j} \left(1 - \sum_s \min\{(P^{b_0})_{is}, (P^{b_0})_{js}\} \right) \leq \delta. \tag{13}$$

Then P has a unique stationary probability π, and

$$|\lambda^{-m} Z_m - \pi_0| \leq \delta^{\lfloor m/b_0 \rfloor} \quad (m \geq 0). \tag{14}$$

In Theorem 5.1, one may set $d = \pi_0$ and replace its middle-error term by

$$[z^H] K(z) \, \frac{1 + z + \cdots + z^{b_0 - 1}}{1 - \delta z^{b_0}}. \tag{15}$$

In Proposition 5.3, use $\delta_d = \delta^{\lfloor J/b_0 \rfloor}$.

Proof. For two probability rows p, q, $\|p - q\|_{\mathrm{TV}} = 1 - \sum_s \min(p_s, q_s)$, where total variation is half the ℓ^1 distance. If a signed row w has zero mass, decompose its positive and negative parts as $w = \theta(p - q)$, $\theta = \|w\|_{\mathrm{TV}}$. For any stochastic Q, write

$$pQ - qQ = \sum_{i,j} p_i q_j (Q_{i,\cdot} - Q_{j,\cdot}).$$

The triangle inequality proves contraction by the maximum pairwise row distance of Q. With $Q = P^{b_0}$ that distance is at most δ. A stationary probability exists: any limit of a subsequence of the Cesaro means of $e_0 P^j$ is stationary, by compactness and telescoping. The strict contraction makes two stationary probabilities equal. Ordinary stochastic multiplication does not increase total variation, so $\|e_0 P^m - \pi\|_{\mathrm{TV}} \leq \delta^{\lfloor m/b_0 \rfloor}$. Its single-coordinate error bounds $|(P^m)_{00} - \pi_0|$. Diagonal similarity leaves that diagonal entry unchanged, giving (14). Finally

$$\sum_{m \geq 0} \delta^{\lfloor m/b_0 \rfloor} z^m = (1 + z + \cdots + z^{b_0 - 1})(1 - \delta z^{b_0})^{-1};$$

insert this coefficientwise majorant in the proof of Theorem 5.1. □

A positive power of P supplies an overlap margin, but a displayed floating-point minimum is not a rigorous margin. The Perron pair, row normalization and entry enclosures must all be certified. If the computed overlap is exactly zero, one may use any positive upper $\delta < 1$ or keep its finite-support remainder directly. At zero winding fugacity the source instead uses its empty-input limit and the strict nonempty-block radius; the positive-power argument should not be imposed on a reducible zero-fugacity row. Similarly, an entrywise inequality $Aw \leq aw$, $w > 0$, implies $\|A^j\|_{w,\infty} \leq a^j$ and supplies $C_A = 1$ in that weighted norm. Bounds for X, endpoints and caps must use the same norm. All these are finite sufficient tests, not automatic uniform-in-N estimates.

8 Confluent Poles Are Part of the Cap Moment

At generic common twist, the source's trace is a finite functional on the root Laurent algebra $\mathcal{A} = \mathbb{C}[p_1^{\pm 1}, \dots, p_N^{\pm 1}]$:

$$\tau_0(f) = \sum_{\nu} \sum_{|\alpha| \le m_\nu} c_{\nu,\alpha}\, \partial^\alpha f(p^{(\nu)}).$$

This is obtained by continuing the independent-twist trace, not by discarding roots at which the ordinary Jacobian vanishes. The source also proves annihilation of every Laurent multiple of the cleared TQ-relation coefficients. Vanishing of those coefficients only at the support points would not be enough to remove their derivative contributions.

Let $F = P(u)/d_{\mathrm{full}}(u)$, where the scalar denominator is regular and nonzero, and fix one row argument u. The exact all-row identity is

$$\Lambda T^{K^{-1}}(u)^H \Psi = \tau_0(F^H)/\tau_0(1). \tag{16}$$

The normalizer $\tau_0(1)$ is fixed by the empty moment and is not an arbitrary Euclidean norm. At a regular isolated selected root the contribution is WF^H/J, with all root labelings accounted for. The source identifies its physical coefficient only after the fixed-N annular gap and cap-end argument.

Lemma 8.1 (Exact jet power rule)**.** *Let $\mathcal{B}$ be a finite-dimensional local commutative algebra over $\mathbb{C}$, let $\mathfrak{m}^s = 0$, and let $F = \zeta + \eta$, $\eta \in \mathfrak{m}$. For every linear functional ℓ and integer $H \ge 0$,*

$$\ell(F^H) = \sum_{r=0}^{\min(H,s-1)} \binom{H}{r} \zeta^{H-r} \ell(\eta^r). \tag{17}$$

For $\zeta \neq 0$ it is ζ^H times a polynomial in H of degree at most $s-1$; for $\zeta = 0$ it vanishes for $H \ge s$.

Proof. Expand by the finite binomial theorem in $\mathcal{B}$. Every term with $r \ge s$ is zero. Apply ℓ. □

This elementary identity explains the source's exponential-polynomial height expansion. It is not a new spectral theorem. It also gives an exact way to test jets without subtracting close ordinary residues. For instance, for $H \ge 1$,

$$\lim_{\epsilon \to 0} \frac{(\zeta+\epsilon)^H - (\zeta-\epsilon)^H}{2\epsilon} = H\zeta^{H-1}.$$

Replacing the two residues by their coincident point value would erase this entire term. For $H \ge 3$, binomial expansion gives the useful error bound

$$\left| \frac{(\zeta+\epsilon)^H - (\zeta-\epsilon)^H}{2\epsilon} - H\zeta^{H-1} \right| \le \binom{H}{3} |\epsilon|^2 (|\zeta| + |\epsilon|)^{H-3}. \tag{18}$$

Indeed sum the odd terms $r \ge 3$ and use $\binom{H}{r} \le \binom{H}{3}\binom{H-3}{r-3}$. The exact error is zero for $H = 1, 2$. The height-dependent bound warns against interchanging a numerical root-coalescence limit and an arbitrarily large-height limit.

If $|\zeta| < \lambda$, each finite polynomial times $(\zeta/\lambda)^H$ vanishes. But a competing term with modulus at least λ, or a nonzero higher jet at λ, cannot be dismissed from this algebra alone. The source excludes these for the selected scalar by the full annular pole argument and nonzero ordinary coefficient, then continues to $K = i, 1$. Those twists may be torically resonant for a generic roots-only description. We retain the analytic continuation; we do not apply the generic toric exclusion there by fiat.

9 Physical Coordinates and Limit Order

Fresh original private readings provide comparison principles, not a substitute proof of the honeycomb model. The physical-metric chapter [4] proves $G = J^*J$ for an injective synthesis map, and the exact full-residual decomposition into projected and omitted components. Its spectral companion [5] supplies an explicit nested graph-core compression with a spurious spectral value. The private spectral gate [6] requires a full tail resolvent and a uniform contour correction before promoting a finite count. These observations apply here as safeguards:

1. A genuine change of basis S sends M to $S^{-1}MS$, a right cap to $S^{-1}\Psi$, and a left cap to ΛS. The scalar $\Lambda M^H \Psi$ is unchanged. Keeping old caps after changing the row matrix changes the problem.

2. In a Hilbert realization, coefficient norms must carry J^*J. In the diagram bounds we instead declare a positive weighted coefficient norm and include its conversion constants. Neither choice converts the source's bilinear cup pairing into positivity.

3. Every fixed-N annular matrix is finite. The separate issue as $N \to \infty$ is quantitative uniformity. An infinite operator claim would additionally require specified embeddings, domains and tail control; no such claim is made here.

The original local-jet derivation in TR117 [7] retains product remainders, coefficient derivatives, moving-endpoint terms, and inverse amplification. Its exact polynomial model is distinguished from an infinite inverse. Our use is analogous but different: the finite local algebra in Lemma 8.1 is exact, while the infinite height sum is justified by $\alpha, t < 1$. We do not assert that this paper's residue identities follow from TR117, or rename its Taylor method as the source Bethe construction. TR250 was located and prehashed but was not needed for a fresh proof passage in this report; it is not counted as read evidence.

Two elementary examples make the risks concrete. With

$$T = \begin{pmatrix} 1 & b \\ 0 & r \end{pmatrix}, \quad 0 < r < 1, \quad \Lambda = (1, 0), \quad \Psi = (1, 1)^t,$$

one has $\Lambda\Psi = 1$, but

$$\Lambda T^H \Psi = 1 + b(1 - r^H)/(1 - r), \qquad \Lambda\Pi_1\Psi = 1 + b/(1 - r).$$

The eigenvalue 1 alone does not determine the amplitude. For the family $T_N = \mathrm{diag}(1, 1 - 1/N)$, paired to sum both diagonal contributions,

$$\lim_{H\to\infty} \left(1 + (1 - 1/N)^H\right) = 1 \quad (N \text{ fixed}),$$

whereas at $H = N$ the expression tends to $1 + \mathrm{e}^{-1}$. Thus our fixed-N certificate is not permission to choose $H = N$ in the source amplitude limit.

10 The Chord Normalization Is Not a Cap Normalization

At winding fugacity 2, the source turns the cap scalar into a positive two-tip gas. The seam twists are $i, 1, -i$. For a counterclockwise contractible loop or a left-oriented winding loop, the source

phase calculation gives the orientation sum

$$2\,\mathrm{Re}\big(i(-i)^{c_0+c_H}\big) = \begin{cases} 2, & c_0 + c_H = 1, \\ 0, & c_0 + c_H = 0 \text{ or } 2. \end{cases}$$

This physical positivity, together with topology and height moments, is necessary for the cylinder-to-plane comparison. An absolute matrix norm cannot replace it.

For a fixed macroscopic cutoff $0 < \delta \le 1/50$, let

$$Z_R(\delta) = \sum_{\gamma:\,\mathrm{diam}\,\gamma \ge \delta R} x^{m(\gamma)}, \qquad M_R(\delta) = \sum_{\gamma:\,\mathrm{diam}\,\gamma \ge \delta R} m(\gamma) x^{m(\gamma)}.$$

These sums run over ordered boundary-port chords of the triangle, of all lengths. The source proves its assertions through

$$Z_R(\delta) \asymp_\delta R^{3/4}, \qquad M_R(\delta) = R^{25/12+o(1)}, \qquad \mathbb{E}m = M_R(\delta)/Z_R(\delta).$$

Neither Z_R nor M_R equals the cylinder cap normalizer. The interior-contact identity is $m = 1 + \#\{\text{interior contacts}\}$, not $m = \#\{\text{contacts}\}$. The exact slit relation is $T_e = 2Z_e + \sqrt{2}L_e$; the marked loop has bare weight $x^{|P|}$, the surrounding loops weight $2x^{|P|}$. These constants and the positive cosine comparison must be retained.

Deleting port legs changes weight by a factor x, changes length from m to $m-1$, and can change the diameter by a bounded amount. The source compares the two fixed cutoffs $1/100, 1/50$ with extension multiplicity at most 9. Its centre-walk mean is then $M_R^\circ/Z_R^\circ - 1$. This is not a theorem about uniformly sampled fixed-length walks, fixed endpoints, or a typical-length lower bound. Our cap certificate does not strengthen any of those laws.

11 Comparison, Costs, and Further Work

Demonstrated increment. The source already proves a fixed-N exponentially small cap-amplitude remainder by absorbing a polynomial into a slightly larger exponential base. Our refinement preserves the actual mark budget, improves that budget from N to $\lfloor N/2 \rfloor$, and makes all three truncation errors explicit. The unequal-depth formula can exploit a cheaper or smaller cap on one side. Its purpose is certified absolute accuracy, not a new amplitude exponent. The stochastic overlap proposition supplies a classical, fully finite sufficient way to validate the middle input; it is not credited as a new mixing theorem.

Advantages and costs. The tail formula is a finite sum of nonnegative terms, which avoids catastrophic cancellation in subtracting large partial sums. It handles $L < k$ and $L = 0$ without a hidden large-depth assumption. Once constants are certified, coefficients through height H can be formed by ordinary polynomial arithmetic, retaining the separate contributions. This does not account for the potentially large cost of constructing cap states, finding a rigorously enclosed Perron vector, or bounding a full annular power. A crude X_* or poorly scaled norm can make the certificate uninformative. The rational majorant is not asserted to be sharp for actual honeycomb diagrams, and no runtime comparison was performed.

Extensions that require more input. Occupancy-resolved cap sums allow different budgets k_0 to be combined with their actual cap weights. The same proof works for any permanent resource consumed on exceptional rows, but applying it to another diagram model requires a new monotonicity

argument. Parameter-uniform certificates require uniform bounds on $\lambda, a, C_A, X_*, B_\pm, B_C$ and the middle remainder. Complex perturbations require analytic local-weight majorants and a separated contour, not just a real overlap margin. A finite compression of an infinite-state model needs its own tail operator control, as in the private spectral gate.

Useful future work is therefore quite specific: construct one small full annular row with rigorously enclosed trigonometric weights; certify its cap coordinates and overlap; evaluate (10) against exact finite-height enumeration; then investigate whether the resulting constants remain manageable with N. None of these unperformed computations is counted as evidence here. Improving an absolute finite-height enclosure does not automatically improve the planar projection, nesting exponent or all-length chord denominator.

Public literature status. The public source README identifies the September 26, 2026 preprint [1]; publication provenance is not proof verification. The primary journal record of Duminil-Copin and Smirnov [2] establishes the honeycomb connective constant and distinguishes that result from a scaling-limit assertion. Krachun and Panagiotis [3] establish a quantitative sub-ballistic bound for uniformly sampled fixed-length walks; its $O(n/\log n)$ displacement statement is about a different law from the source's critical-fugacity triangle chords. We checked these public abstracts and metadata afresh on October 9, 2026, including the latter's 2026 journal record. Their full proofs were not independently audited here. The search is not an exhaustive priority review, and no independent global validation follows from these citations.

12 Evidence and Contribution Assessment

The owned checker uses exact rational arithmetic throughout. Its parameter cases and assertions are separately counted:

Fixture family	Cases	Assertions
Negative-binomial tails	3192	6384
Noncommuting exceptional-row words	135	270
Signed cap-height enclosures	1664	3328
Unequal cap truncation	729	1458
Stochastic overlap bounds	432	864
Nilpotent jet powers	408	816
Confluent regularization	200	400
Worked amplitude example	1	4
Total	6761	13524

The signed fixtures prevent reliance on cancellation-free physical caps in the analytic enclosure. The word fixtures use genuinely noncommuting matrices. These checks corroborate the proved local identities; they do not check the full honeycomb R-matrix enumeration, global residue theorem, root census or infinite asymptotics. Parent checks are not included.

The read ledger records all 5456 lines in nineteen public text files and 854 freshly displayed private lines in four files. The private intervals are: physical metrics 1–210; spectral limits 1–242; TR-core-pat-1 855–996; TR117 516–775. The last two files were not read in full. TR250 is a prehashed locator only. Every complement is explicit in the ledger. The public and private prehash manifests were persisted before these proof displays; final hashes are rechecked after reading. Unread external proofs and the remainder of the private corpus are not represented as inspected.

The contribution is assessed as **C3**, independently of the catalogue's **L3** applicability. It is a notable local quantitative refinement of the actual cap mechanism, with explicit stopping errors and assumptions. The classical binomial, Perron, overlap-contraction and finite-jet algebra are credited rather than renamed. C4 or C5 would be unjustified: the report does not supply the full physical constants, improve the global powers, establish a uniform infinite spectral limit, or independently certify the source's claimed honeycomb theorem.

References

[1] OpenAI, *Cap-selected amplitudes and triangle chords for honeycomb walks*, September 26, 2026. Pinned public source and citation. All nineteen cached text files read; global arguments remain source inputs except for the local deductions proved here.

[2] H. Duminil-Copin and S. Smirnov, *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$, Annals of Mathematics 175 (2012), 1653–1665. Primary journal record. Abstract and metadata checked; full proof not independently audited.

[3] D. Krachun and C. Panagiotis, *Quantitative sub-ballisticity of self-avoiding walk on the hexagonal lattice*, Annals of Probability 54 (2026), 1109–1125, doi:10.1214/24-AOP1730. Author preprint; Author institution's journal record. Abstract, version and publication metadata checked.

[4] M. Subbarao, private RamCalBooks, volume 21, *A Nonorthogonal Representation Carries a Metric* and subsequent sections, local file `ch01d-physical-metrics.tex`, lines 1–210. Fresh original chapter read; no private content uploaded.

[5] M. Subbarao, private RamCalBooks, volume 21, *A Dense Sequence With a Spurious Spectral Value* and subsequent sections, local file `ch01f-spectral-limits.tex`, lines 1–242. Fresh original chapter read, including residual and inverse limits.

[6] M. Subbarao, private `TR-core-pat-1.tex`, section *Spectral Counts and Tail-Resolvent Promotion*, lines 855–964, and following release discussion through line 996. Selected original proof read; other sections not freshly read.

[7] M. Subbarao, private original foundation report, `TR-rev-1-old-tr117.tex`, lines 516–775. Taylor/jet forward remainders, coefficient and endpoint derivatives, and conditional finite inversion. Selected proof passage only; the report is not asserted to prove the source residue identities.